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Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

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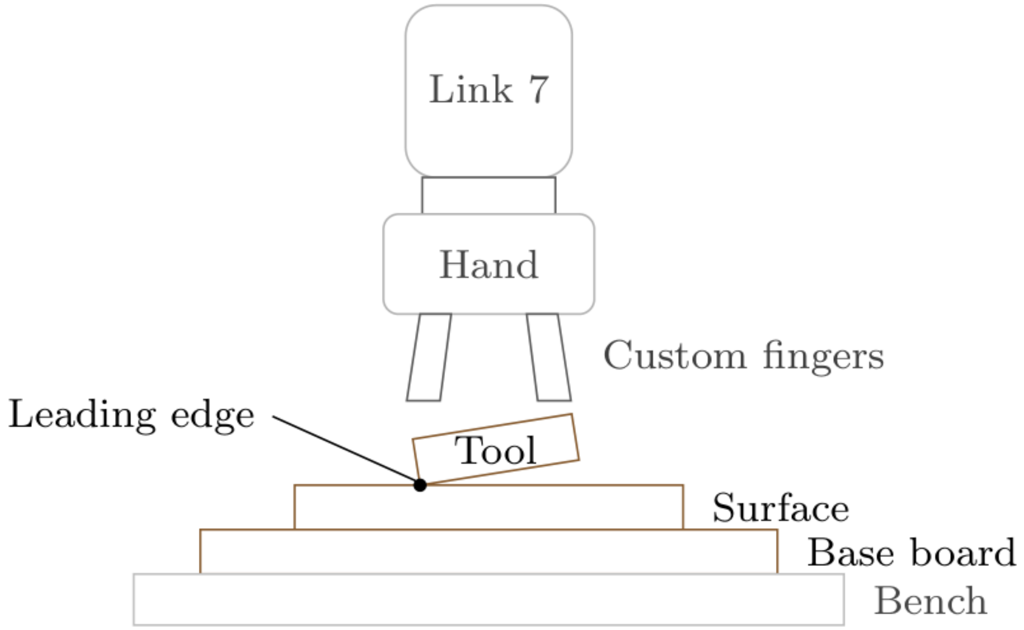
Motivation



- # Problem
  - Tilted tool → edge-first contact
  - Rigid tracking resists corrective rotation
- # Main thesis question
  - **How do Cartesian impedance settings affect passive alignment?**
- # Idea
  - Normal pressing with contact-induced rotation

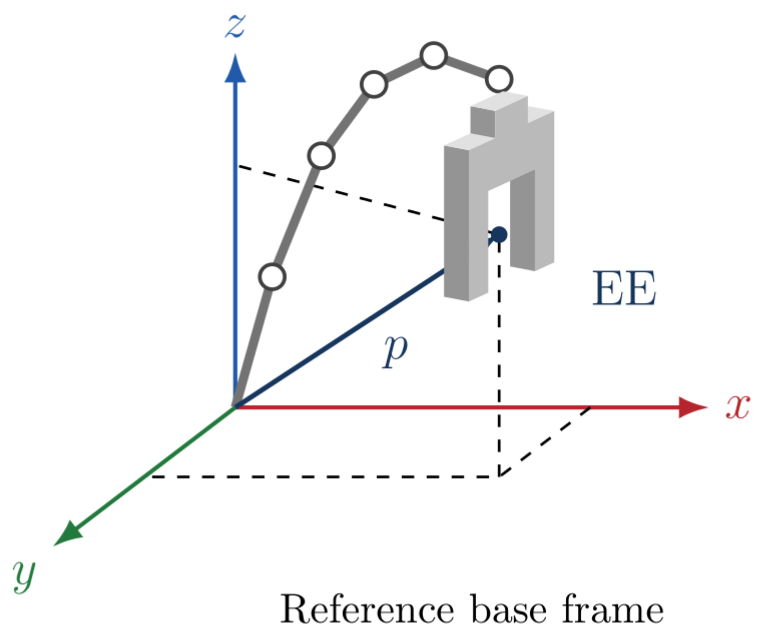
Overview

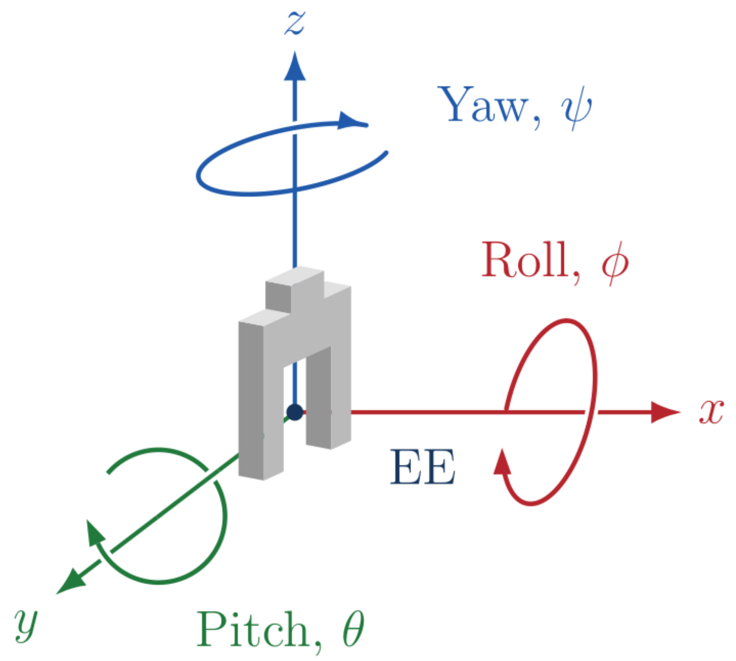
1. Introduction
2. Controller design
3. Experimental methodology
4. Validation
5. Experimental results
6. Conclusion and future work

Cartesian pose

In 3D, the end effector's position and orientation have 6 degrees of freedom.

- Position: 3 translations
- Orientation: 3 rotations





- The EE position depends on the configuration of all 7 joint angles: $p = p(q)$

Cartesian impedance controller

- Position correction by force

$$e_p = p_d - p_{\mathrm{EE}}$$

$$f = K_p e_p + D_p(\dot{p}_d - \dot{p}_{\mathrm{EE}})$$

- Orientation correction by moment

$$e_R = \phi u, \qquad R_d R_{\mathrm{EE}}^\top = R(u, \phi)$$

$$m = K_R e_R + D_R(\omega_d - \omega_{\mathrm{EE}})$$

- Combined Cartesian wrench

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \underbrace{\begin{bmatrix} K_p & \mathbf{0} \\ \mathbf{0} & K_R \end{bmatrix}}_{K_c} \begin{bmatrix} e_p \\ e_R \end{bmatrix} + \underbrace{\begin{bmatrix} D_p & \mathbf{0} \\ \mathbf{0} & D_R \end{bmatrix}}_{D_c} \begin{bmatrix} \dot{p}_d - \dot{p}_{\mathrm{EE}} \\ \omega_d - \omega_{\mathrm{EE}} \end{bmatrix}$$

- Zero off-diagonal blocks: translation and rotation are decoupled.

$K_p,\ K_R$	translational / rotational stiffness	$D_p,\ D_R$	translational / rotational damping
$\dot{p},\ \omega$	linear / angular velocity	$d,\ \mathrm{EE}$	desired / measured end effector

Real-time control

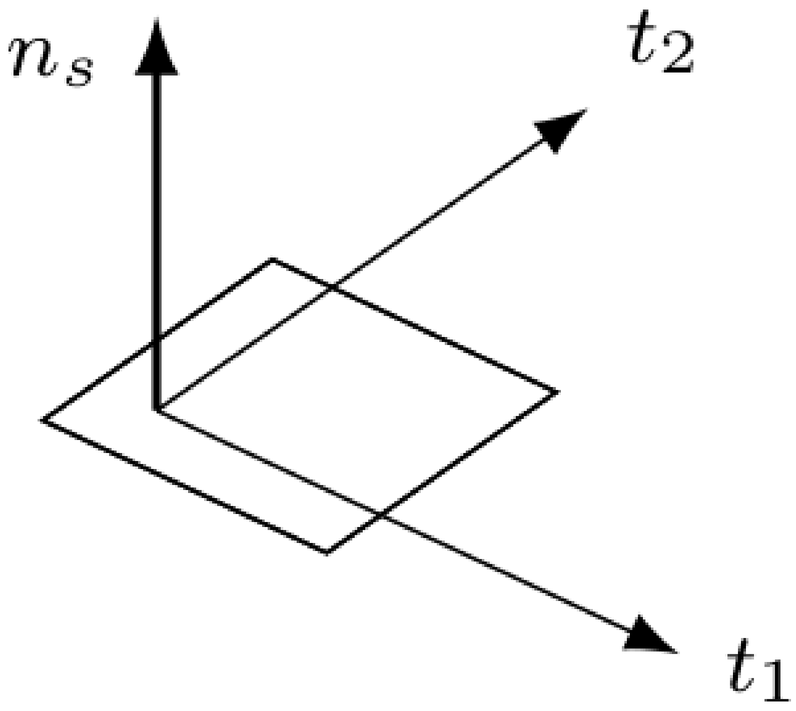
$$\text{Pose error} \longrightarrow \text{Wrench} \longrightarrow \text{Joint torques} \longrightarrow \text{Robot}$$

Measured pose

$$F = \begin{bmatrix} f \\ m \end{bmatrix}, \qquad \tau_{\text{cart}} = J^{\top}(q)F$$

1 kHz control loop · C++ / libfranka

Surface-relative compliance



- n_s outward surface normal
- $t_1,\ t_2$ two perpendicular surface tangents

- Translation
  - Softer normal pressing
  - Stiffer motion along the surface
- Rotation
  - Softer rotations about the tangents
  - Stiffer rotation about the normal

Centre of compliance (CoC)

- Virtual reference point

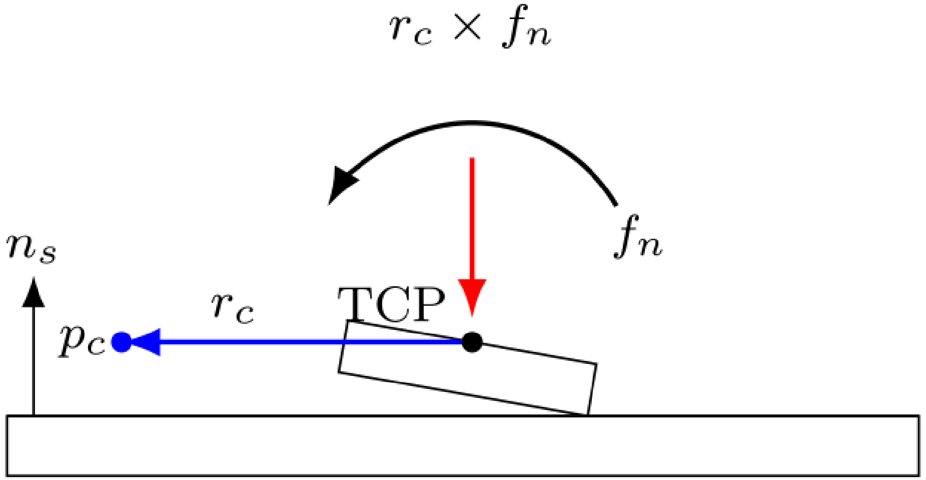
p_c CoC position

r_c Displacement from TCP to CoC

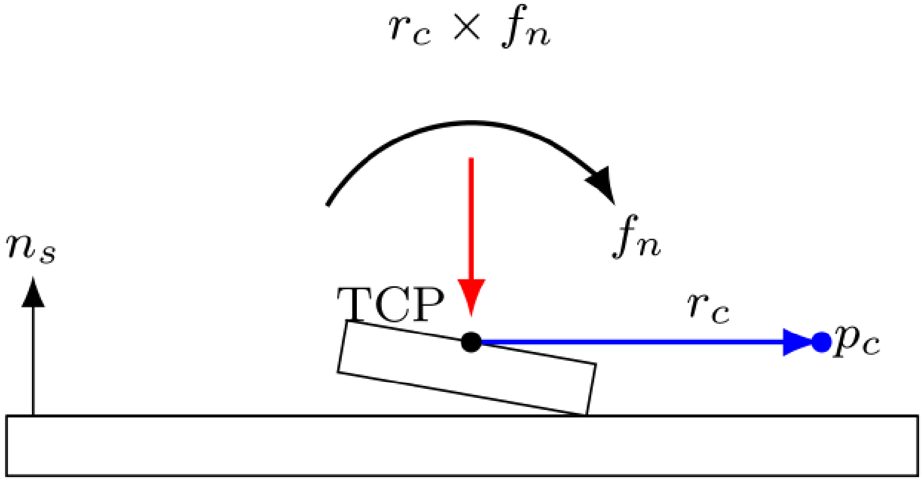
- Commanded TCP moment

$$m = m_R + r_c \times f$$

m_R rotational spring-damper moment



(a) Supporting displacement



(b) Reversed displacement

- Opposite displacement reverses the added moment

Translation–rotation coupling

- Shift from TCP to centre of compliance

$$A = \mathrm{Ad}(r_c), \qquad e_c = Ae$$

Local small-angle model with uncoupled stiffness and damping at the CoC

- Stiffness

$$K_{\mathrm{TCP}} = A^{\top} K_c A$$

- Damping

$$D_{\mathrm{TCP}} = A^{\top} D_c A$$

- A non-zero CoC displacement creates translation–rotation coupling.
- Translational error and velocity can contribute to commanded moment.

Null-space controller

- Redundancy

  7 joints for a 6-DOF Cartesian pose.
  One redundant direction at full rank.

  $$\dim(\ker J) = 7 - 6 = 1$$

- Cartesian task preservation

  Zero instantaneous TCP translation and rotation

  $$J(q)\,\dot{q}_{\text{null}} = 0$$

- Damping τ_d

  Opposes redundant joint velocity.
  Suppresses unwanted joint motion.

- Conditioning τ_σ

  Seeks better Jacobian conditioning.
  Can initiate joint motion.

- Singularity: a configuration where the Jacobian loses rank.

Contact experiment

Tool orientation → Surface approach → Contact establishment → Surface grinding

- Compliant contact settings
- Normal translation

$$K_{p,n} = 350\,\mathrm{N/m}$$

- Rotations about the surface tangents

$$K_{R,t_1} = K_{R,t_2} = 5\,\mathrm{N\,m/rad}$$

- Tangential translation

$$K_{p,t_1} = K_{p,t_2} = 2000\,\mathrm{N/m}$$

- Rotation about the surface normal

$$K_{R,n} = 50\,\mathrm{N\,m/rad}$$

- Controlled comparisons
  - A: TCP baseline for zero, positive and negative entry tilt
  - B: rotational stiffness about t_1
  - C: translational stiffness along t_2
  - D: tangential CoC displacement

- Planar contact with a 40 × 120 mm tool and a fixed entry reference for 5 s

Null-space experiment

- Commanded joint disturbance

$$\tau_{\text{dist}} = J_p(q)^\top f_{\text{dist}}$$

Joint-torque equivalent of a virtual 20 N point force on link 3.

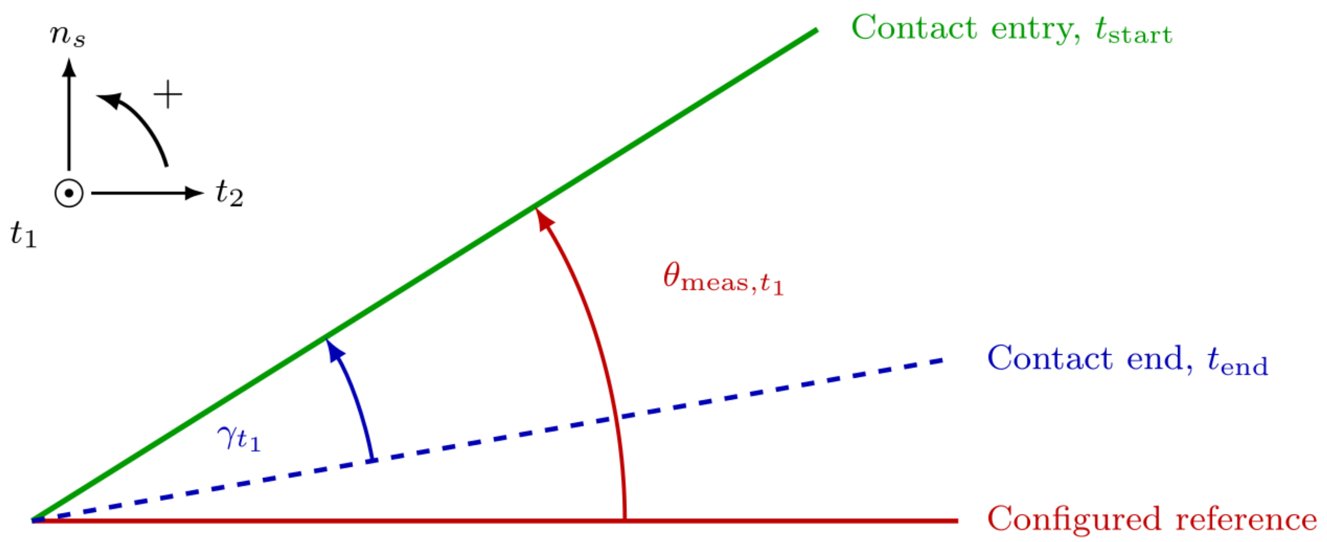
- Cartesian pose hold

$$\|p_d - p_{\text{EE}}\| < 2\,\text{mm}$$

Hold the captured Cartesian pose. Assess TCP position retention.

- Four settings with three trials each
  - No null-space torque
  - Damping: 2 N m s/rad
  - Conditioning: 1.5 and 2.0 N m
  - Compare accumulated motion and net displacement
  - The disturbance excites joint motion while the Cartesian controller holds the TCP.

Angular quantities



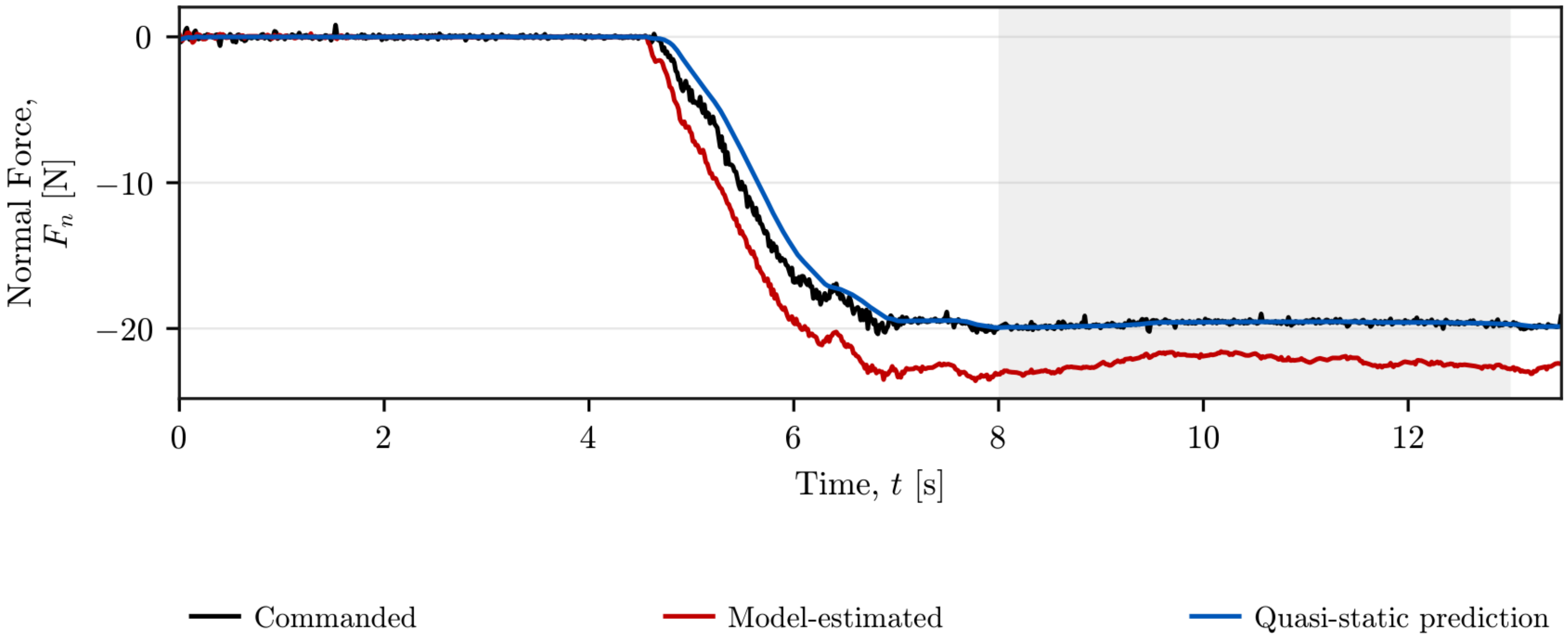
θ_{meas,t_1} Measured angular offset
Relative to the configured surface reference

γ_{t_1} Contact response
Rotation from contact end back to entry orientation

- Aligning response: same sign as the measured angular offset
- Calculated from measured end-effector orientation

Normal-force plausibility assessment

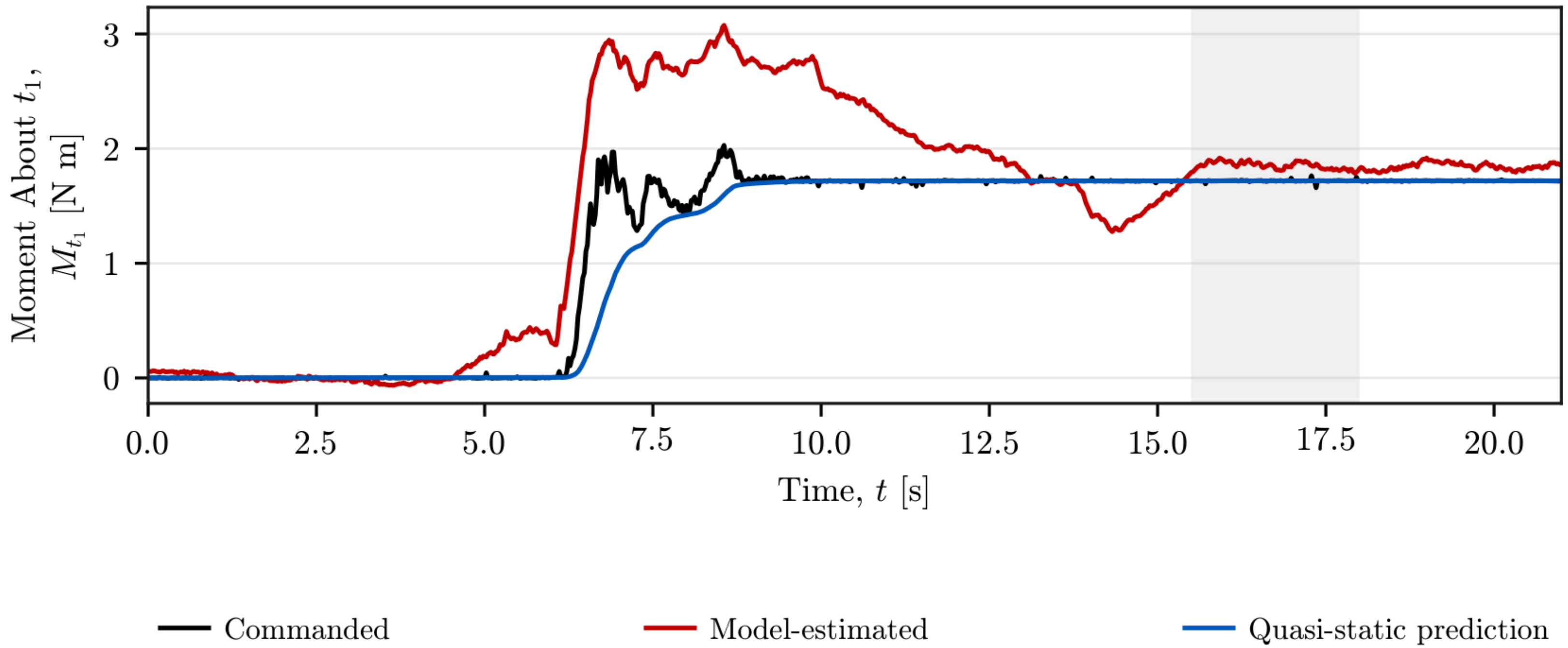
$$F_{n,\mathrm{qs}} = K_{p,n}\, n_s^\top (e_p - e_{p,0})$$



- Spring vs command: 0.05% difference
- Estimate vs command: 13.50% difference

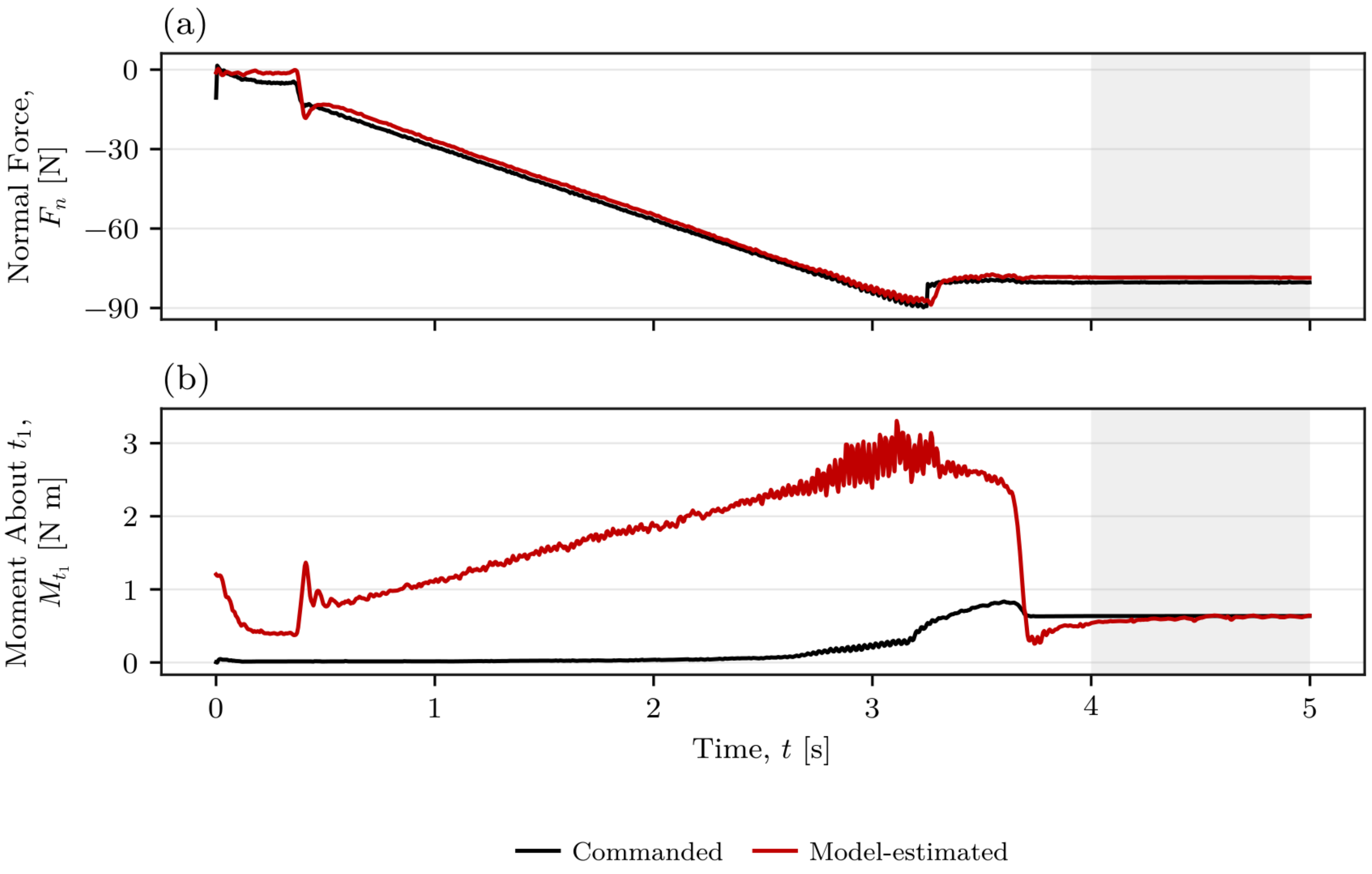
Moment plausibility assessment

$$M_{t_1,\mathrm{qs}} = K_{R,t_1}\, t_1^\top (e_R - e_{R,0})$$



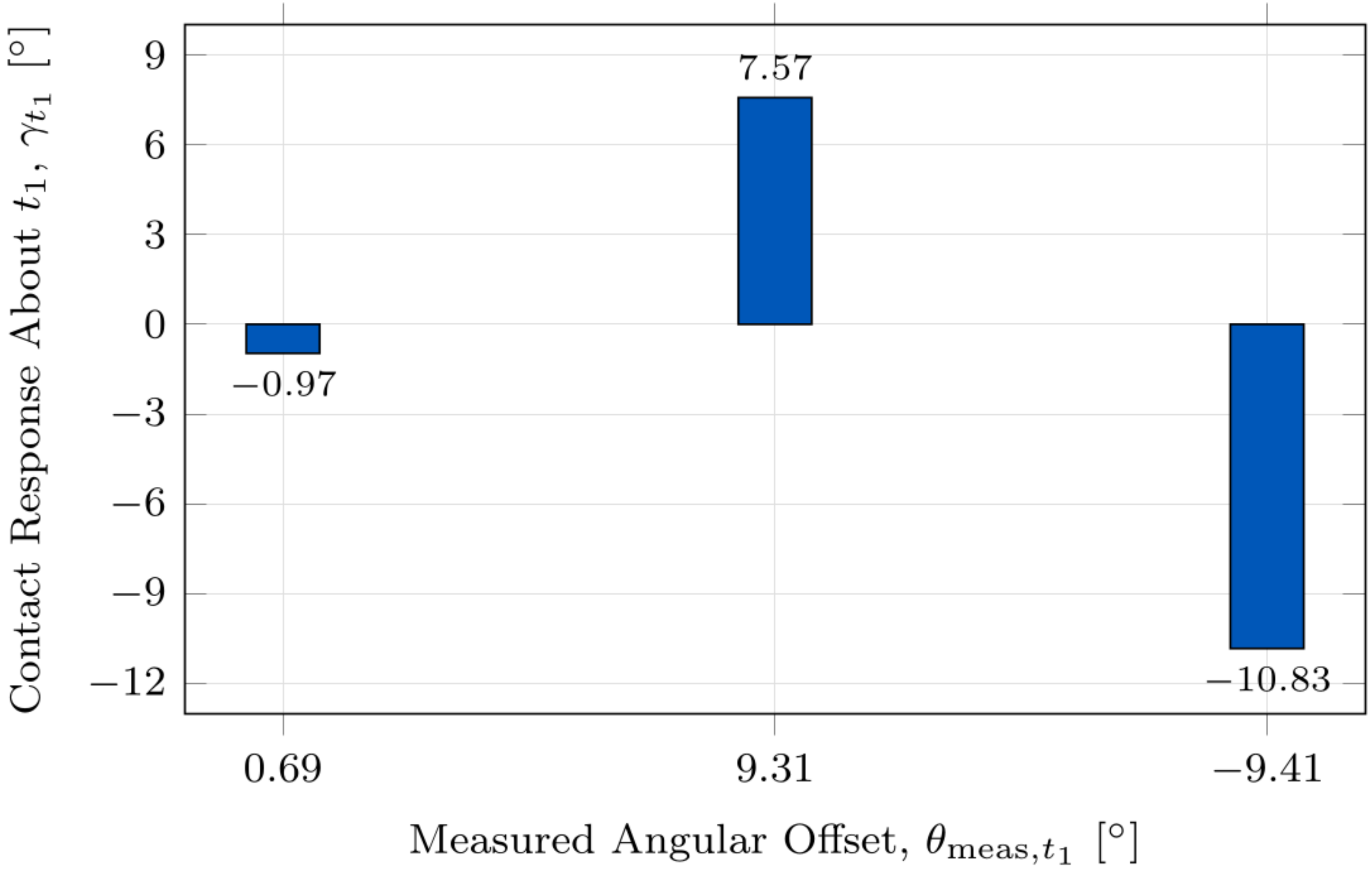
- Spring and commanded mean: 1.719 N m
- Model-estimated mean: 1.852 N m

Commanded and estimated wrench



- Stationary force difference: 2.32%
- Stationary moment difference: 4.87%
- TCP-centred trial with common axes and reference point

Baseline contact response

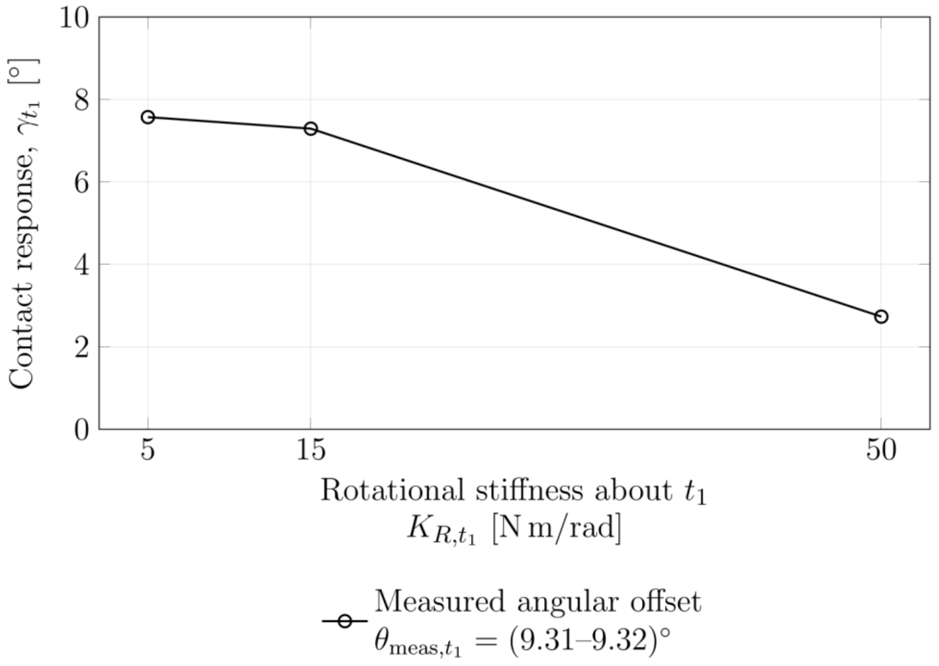


- Centre of compliance at the TCP
- Alignment response for both tilt signs
- Unequal response magnitudes

Effect of stiffness

- Rotational stiffness

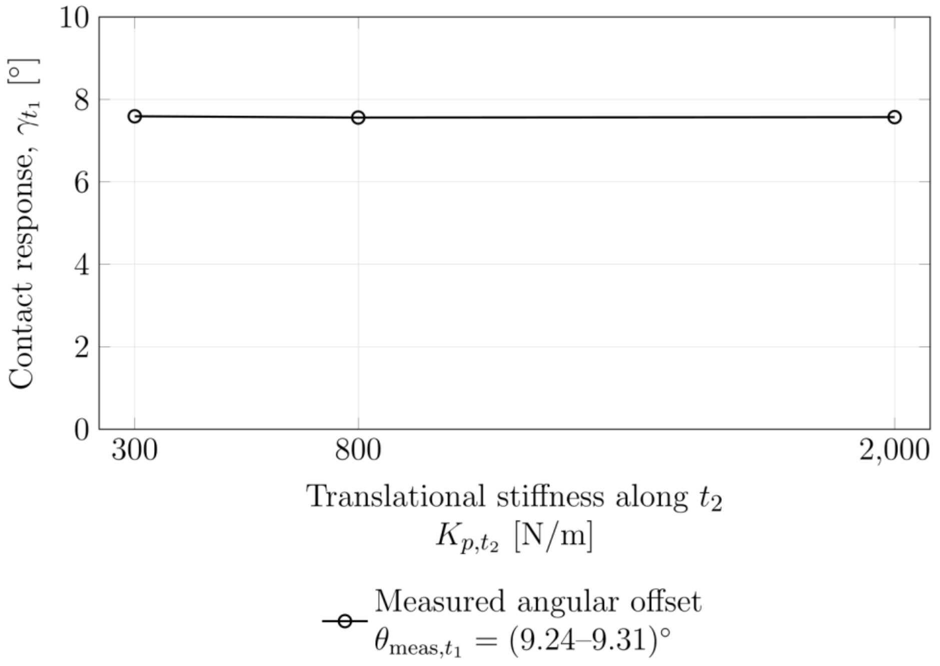
  Resistance to rotation about t_1



- Rotational stiffness: 63.9% lower response

- Translational stiffness

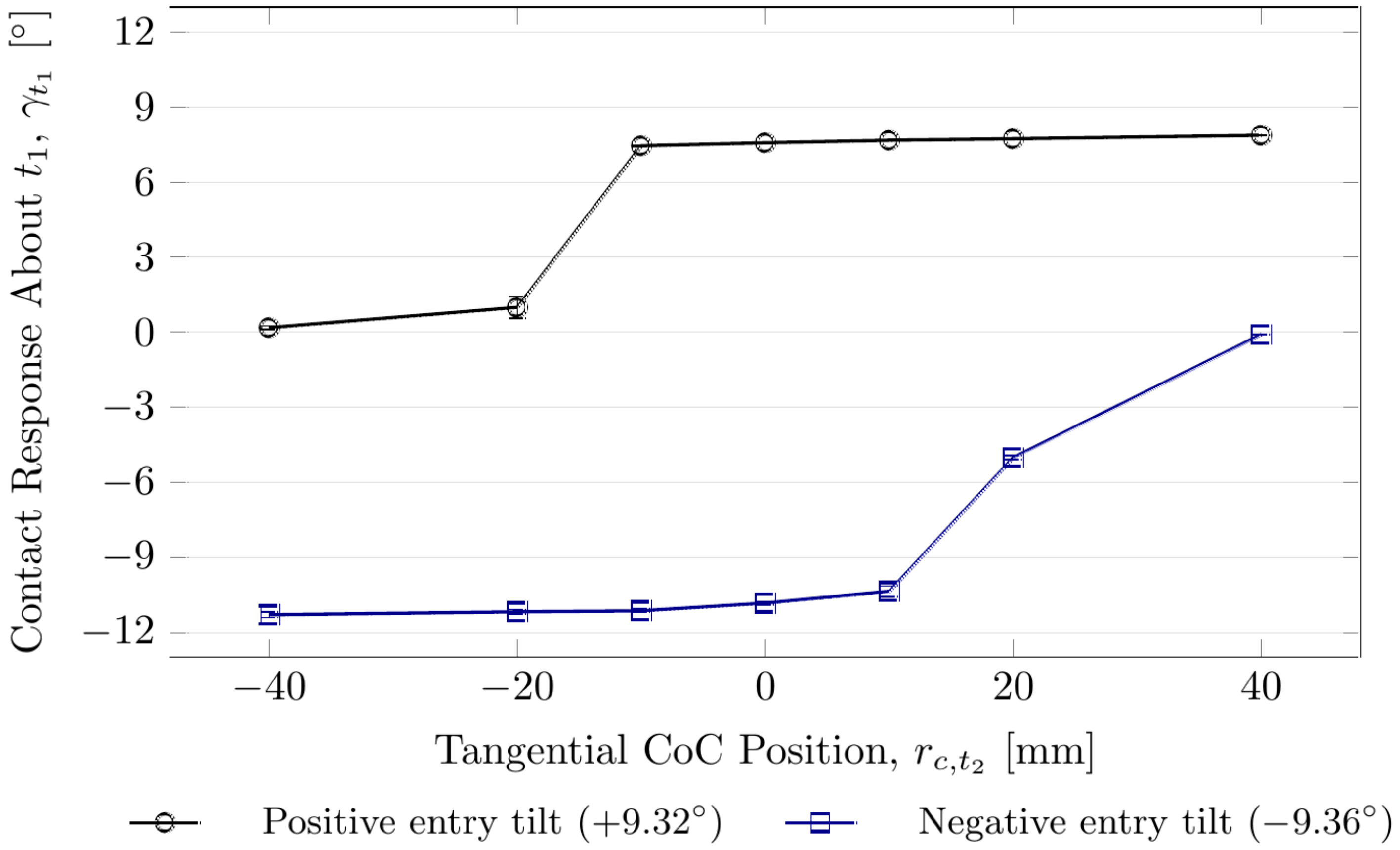
  Resistance to sliding along t_2



- Tangential stiffness: 0.4% response span

Effect of CoC position

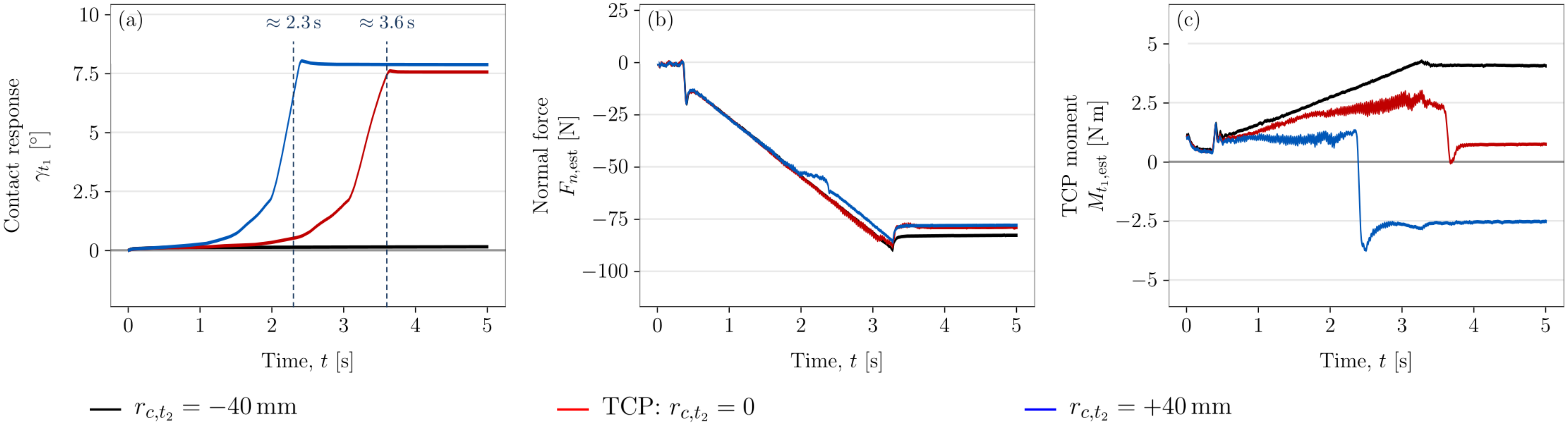
r_{c,t_2} CoC displacement along t_2



Relative to the TCP-centred baseline

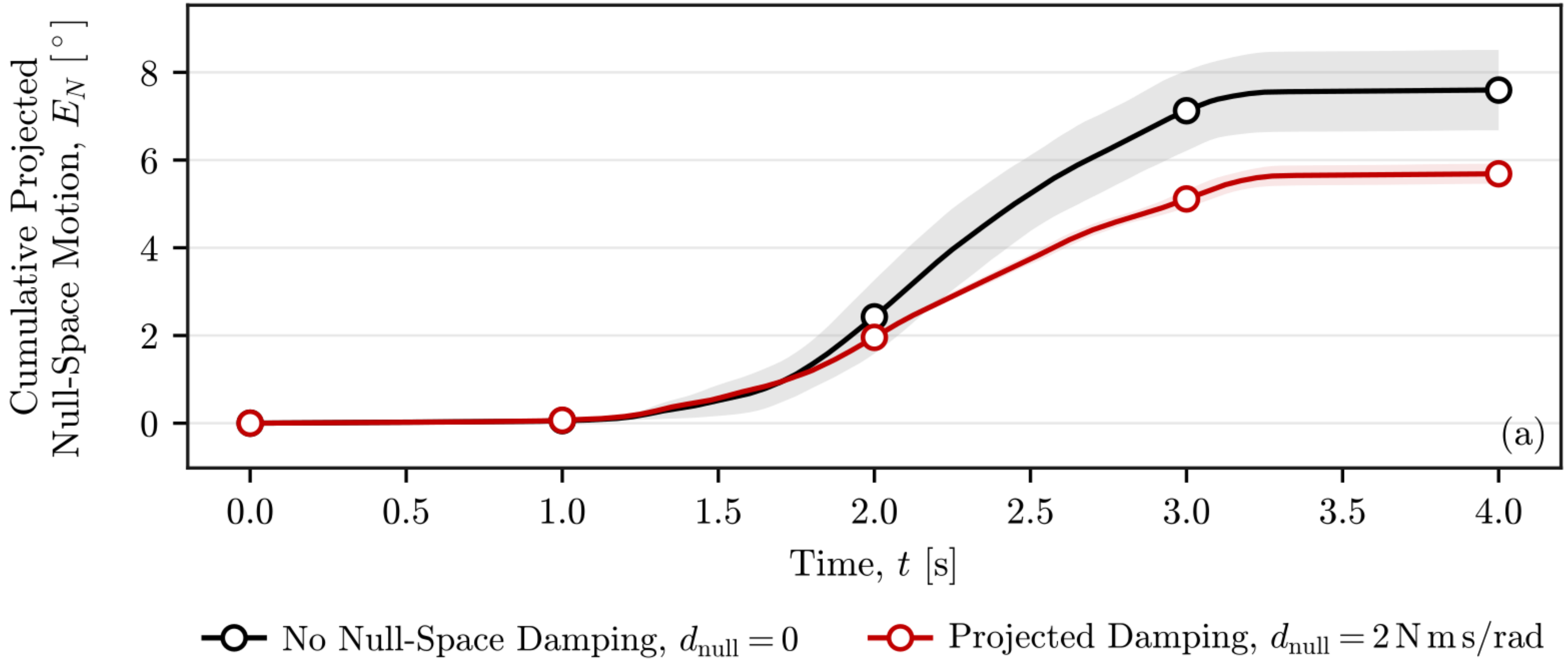
- Supporting displacement: about 4% higher response
- Opposite side: 98–99% lower response
- Required side reverses with the entry tilt

- Contact response
- Normal force
- TCP moment



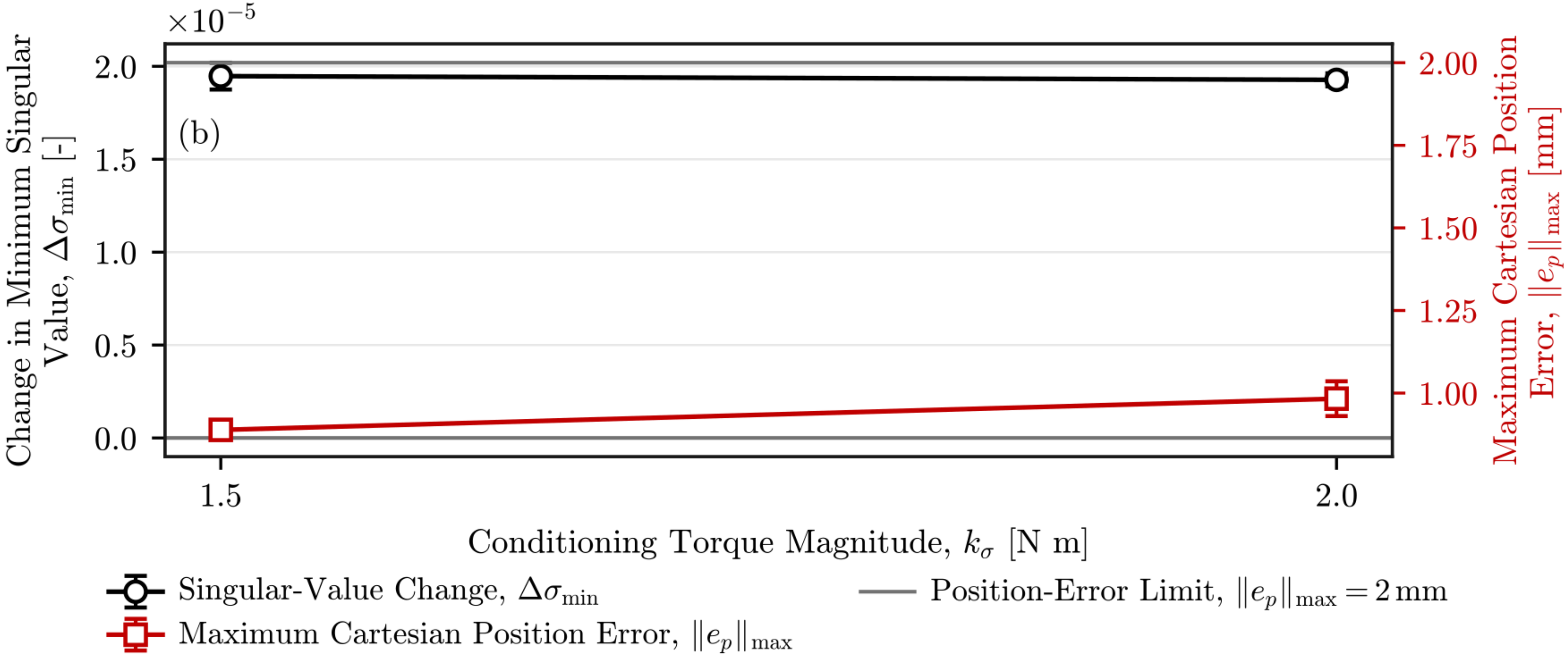
- Reversing CoC displacement reverses the interaction moment
- Supporting displacement: about 1.3 s earlier in these representative traces
- Final normal force (approx.): −83 N (−40 mm), −79 N (TCP), −78 N (+40 mm)

Null-space results



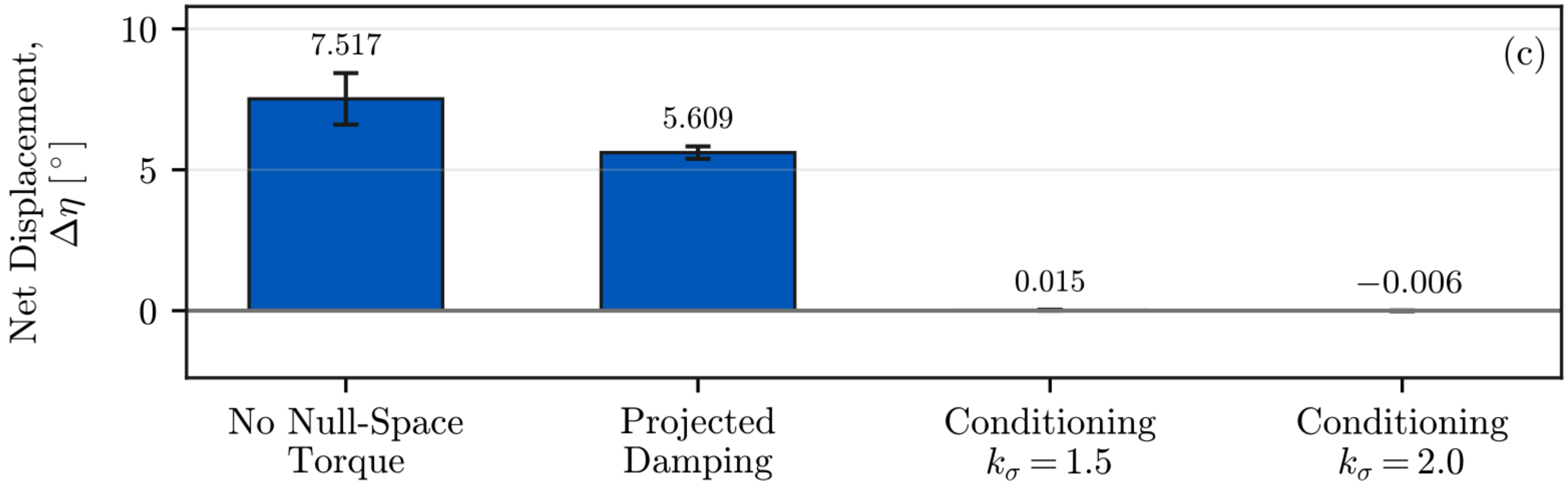
- Damping reduced cumulative motion from 7.596° to 5.687° (25%).
- Identical commanded disturbance across three trials per setting

Conditioning and Cartesian retention



- Both settings improved the local conditioning indicator.
- Maximum TCP error across all trials: 1.304 mm (criterion: 2 mm).

Net null-space displacement



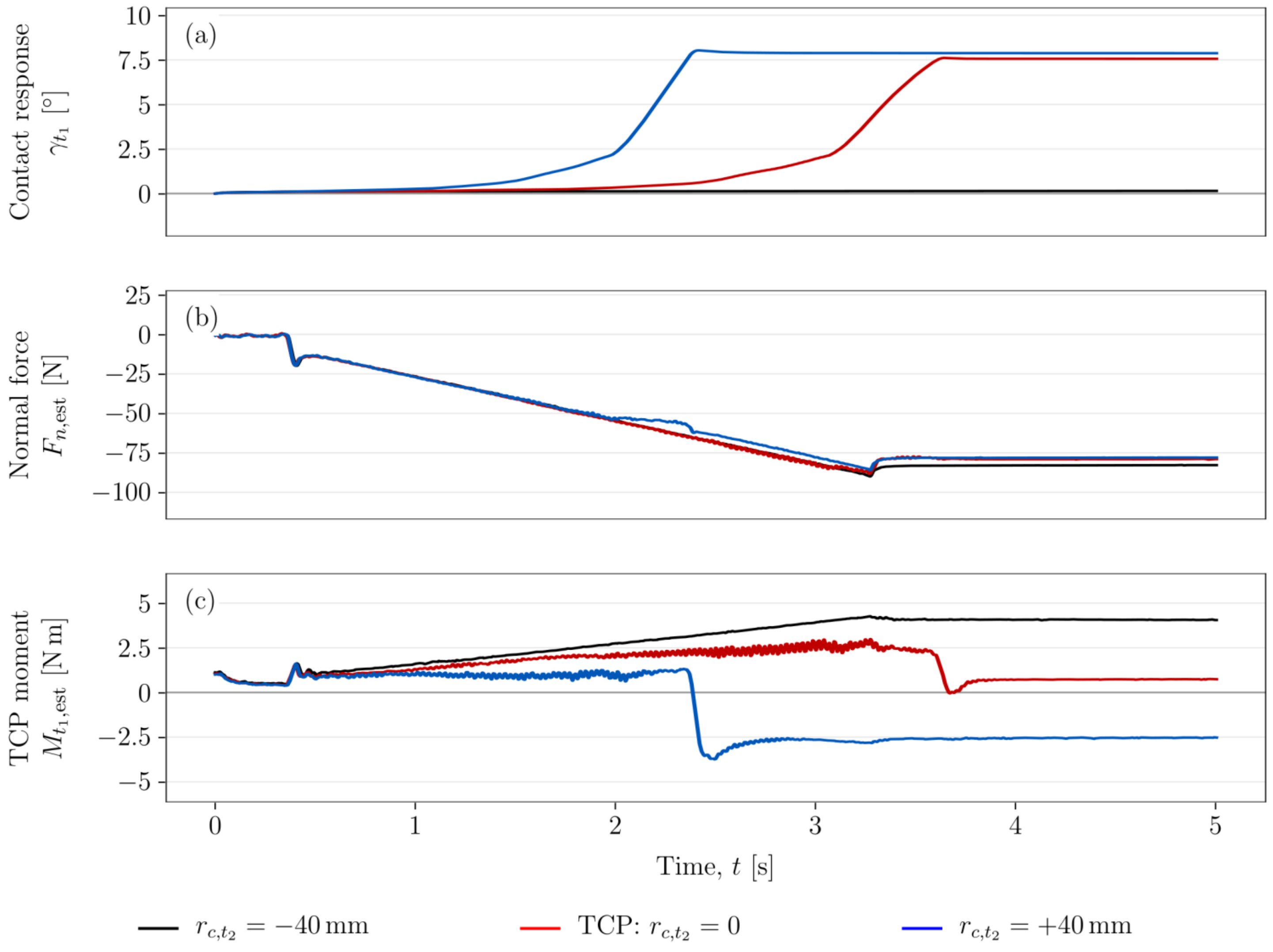
- Conditioning reduced the net displacement to nearly zero.
- Small net displacement can coexist with back-and-forth motion.

Conclusion

- Passive alignment through Cartesian impedance
- Higher rotational stiffness reduced contact rotation
- Changing tangential stiffness had almost no effect on contact rotation
- CoC displacement supports or opposes alignment
- Damping reduced motion and conditioning countered the tested disturbance

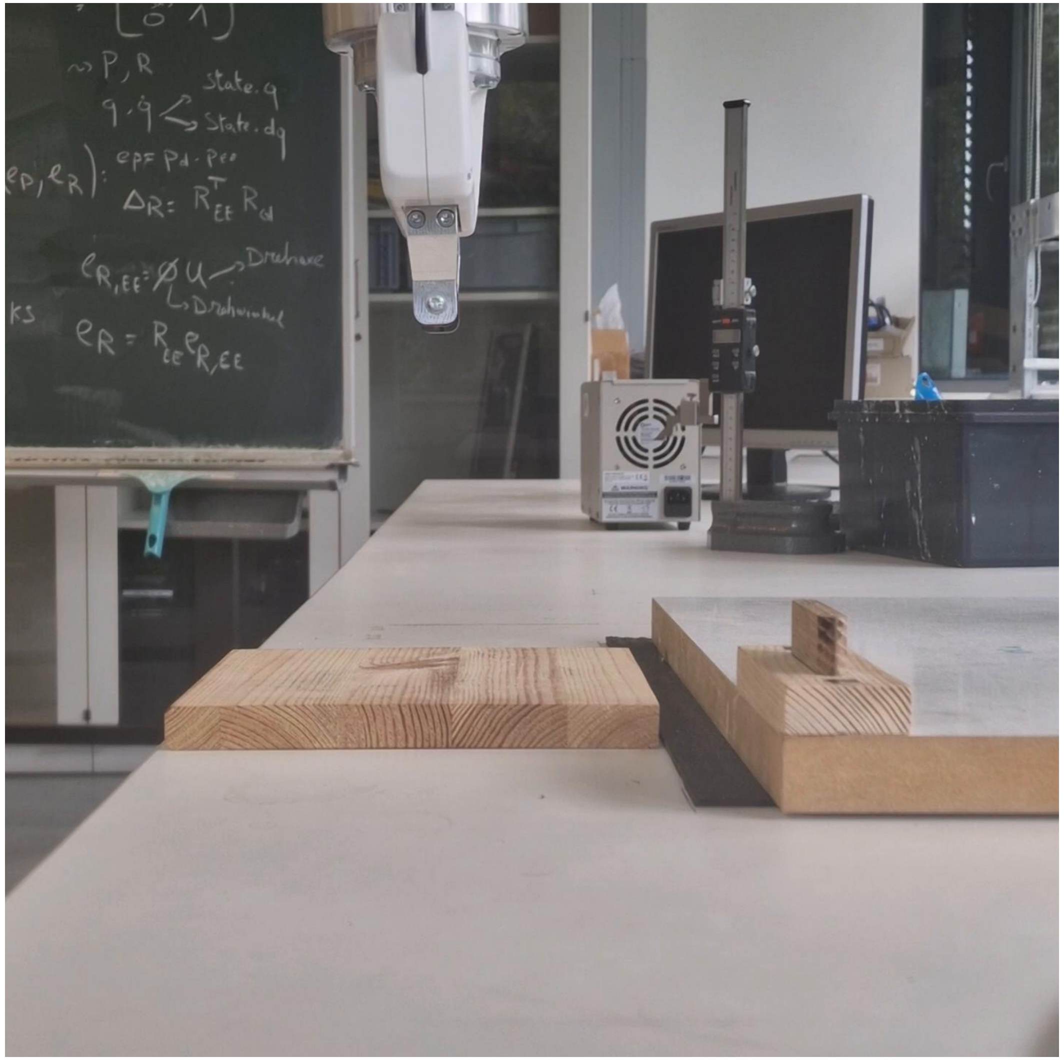
- Future work
  - More rigid tool mount
  - Adaptive CoC selection and return to the TCP after alignment

Contact response, normal force and moment



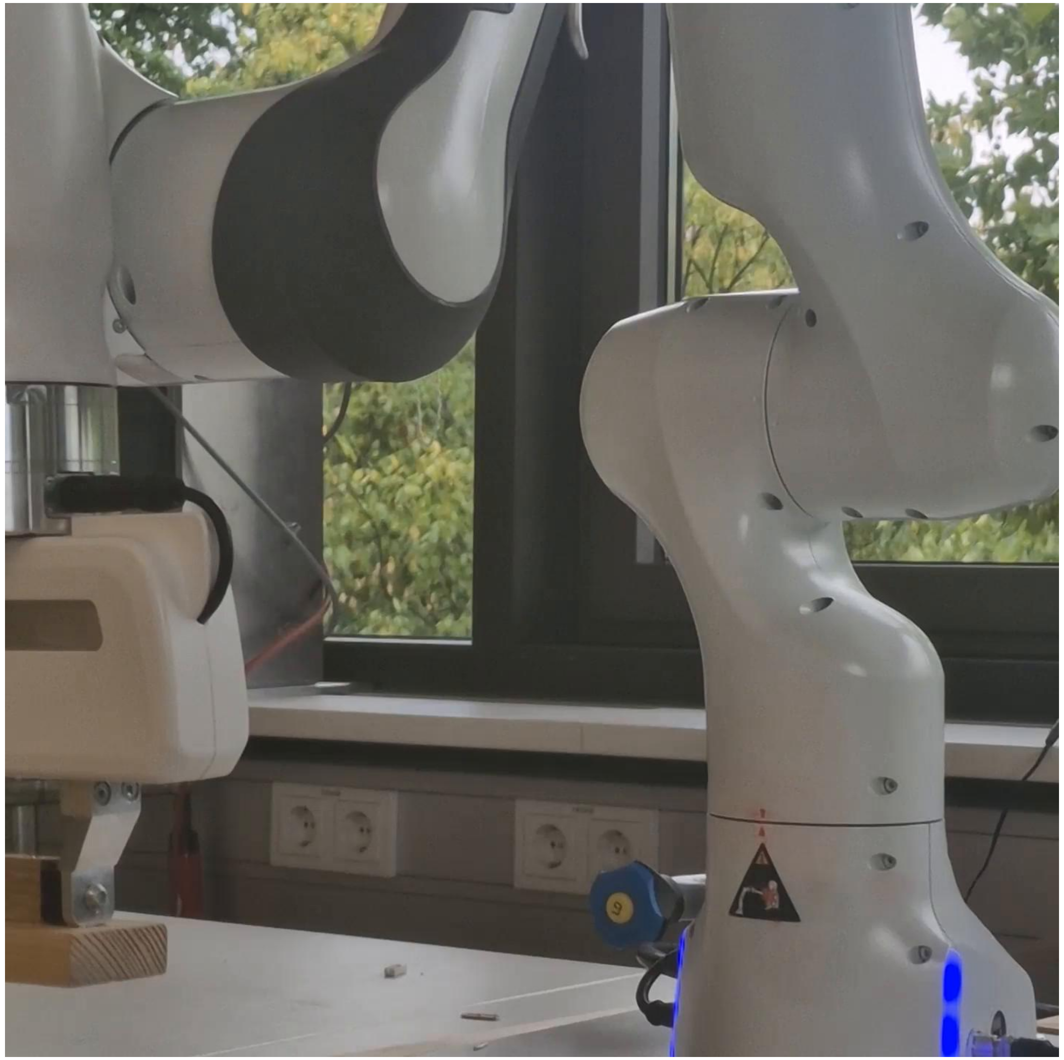
- Representative traces
  Positive entry tilt
- Similar final normal force
- Moment reverses with CoC displacement
- Force and moment: model-based robot estimates

Contact demonstration



Pre-grinding hold · fixed desired orientation · CoC at TCP

Disturbance demonstration



No null-space torque → Damping → Conditioning